HOMEWORK 5.3 · UNIT 5 · NC.8.G.7 · NC.8.G.6 · NC.8.NS.2

Pythagorean Theorem Day 3

Jan 8 / Jan 11 · Targets: G.7-4, G.7-3, G.6-4, NS.2-2, G.7-6, G.7-5, G.7-8, G.7-7

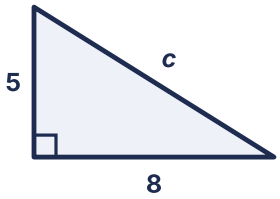
Name _____ Date _____ Block _____

Remember: 1) Draw it and find the right angle. 2) $a^2 + b^2 = c^2$ with the hypotenuse c alone. 3) Take the square root **last**. Not a perfect square? Write the exact answer ($\sqrt{52}$) or round to the nearest tenth. Example: legs 4 and 6 → 16 + 36 = 52 → $c = \sqrt{52} \approx 7.2$.

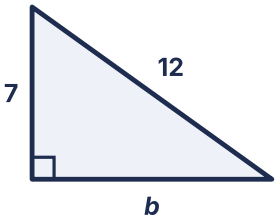
SCORE 3.0 Approaching · the basics

Targets: G.7-4, G.7-3, G.6-4, NS.2-2

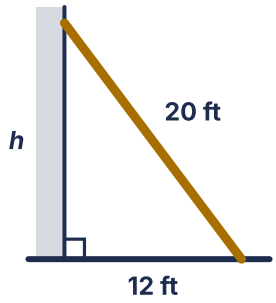
1 Find c. Give the exact answer and round to the nearest tenth. G.7-4



2 Find b. Give the exact answer and round to the nearest tenth. G.7-4



3 A 20-ft ladder leans on a wall. Its base is 12 ft from the wall. How high does it reach? G.7-3



4 Is 9, 12, 16 a right triangle? REVIEW 5.2 G.6-4

Yes No

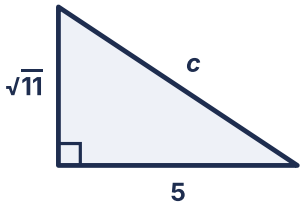
Why? _____

5 $\sqrt{30}$ is between which two whole numbers? REVIEW 1.4 NS.2-2

(A) 5 and 6 (B) 4 and 5 (C) 6 and 7 (D) 15 and 16

SCORE 4.0 Proficient · on target Targets: G.7-6, G.7-5

6 Find c. G.7-6



7 **Nia** made a mistake. Circle it. Explain it. Fix it. G.7-5

NIA'S WORK
Hypotenuse 10, one leg 6.
$10^2 + 6^2 = 136$
other leg = $\sqrt{136} \approx 11.7$

8 A ship sails 9 km west, then 12 km south. How far is it from the port? G.7-5

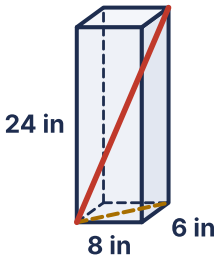
Ⓐ 15 km Ⓑ 21 km Ⓒ 225 km Ⓓ 3 km

SCORE 5.0 Mastery · go deeper Targets: G.7-8, G.7-7

9 A right triangle has legs 3 and 6. Find the hypotenuse as a simplified radical. Explain. G.7-8

c = _____ . The biggest perfect square in it is _____ , so _____ .

10 A box is 8 in long, 6 in wide and 24 in tall. How long is the red diagonal from a bottom corner to the opposite top corner? Explain. G.7-7



First I find _____ . Then _____ .